

Tayaba Abbasi

Machine Learning Scientist | Computer Vision & Spatial-Temporal Modelling

☎ (778)-637-2220 | 📍 Vancouver, BC (Open to relocation) | [G Google Scholar](#) | [in LinkedIn](#) | [@ «tayabaabb@gmail.com»](#)

PROFESSIONAL SUMMARY

Machine Learning Scientist with six years of applied experience building and deploying deep learning systems, spanning computer vision, spatial-temporal sensor analytics, and time-series forecasting. Currently embedded at the **Alberta Machine Intelligence Institute (Amii)** through the Level Up program, advising early-stage AI startups on architecture and applied research. Designs and fine-tunes neural networks from first principles in **PyTorch** and **TensorFlow** CNNs, GANs, encoder-decoder Transformers with a track record of shipping models against noisy, real-world sensor and clinical data in regulated environments across healthcare, oil & gas, and financial domains. Published in **IEEE** and **Springer** with **341+ citations**.

EDUCATION

Simon Fraser University

Master of Science in Professional Computer Science

Relevant Coursework: Data Science, Machine Learning, Big Data, Natural Language Processing, Predictive Modelling

SKILLS

Programming Languages: Python, SQL, R

AI Frameworks: PyTorch, Hugging Face Transformers, Sklearn

ML Libraries: NumPy, Pandas, Scikit-learn, Statsmodels, OpenCV, Matplotlib

Tools: linux, Git version control

MLOps & Deployment: Docker, MLflow, Azure ML, GCP, AWS, Databricks, Apache Airflow, Spark, Kafka

Gen AI & Agents: LangChain, RAG, OpenAI SDK, LLM evaluation & orchestration

Spatial-Temporal & Signal Analytics: Time-series modelling, anomaly detection, real-time sensor-stream ingestion, predictive maintenance, forecasting under distribution shift

MACHINE LEARNING EXPERIENCE

Machine Learning Associate

April 2026 - Present

Alberta Machine Intelligence Institute (Amii)

- Conducting independent pre and post-session research on assigned ML topics, including **LLMs, reinforcement learning, and computer vision** to prepare actionable guidance for startup clients.
- Summarizing and synthesizing cutting-edge AI research papers to surface key insights relevant to each startup's technical challenges and product roadmap.
- Collaborating within **Amii's Level Up CTO Coaching Program** to provide continuous ML advisement to 3–8 early-stage Canadian AI startups, supporting their technical decision making and capability growth.
- Developing **technical reports** after each coaching session, distilling complex ML concepts and research findings into clear, client-ready documentation elucidating core machine learning concepts to enhance understanding

Machine Learning Research Associate

Feb 2022–Feb 2023

UAE University

- Designed and implemented a **ulti-modal deep learning pipeline** in PyTorch (**CycleGAN, U-Net, CNN classifiers**) combining image segmentation with classification, achieving generalization across 2 unseen clinical datasets and cutting domain-shift error by **3%**.
- Built a **graph-based feature representation framework** for multi-modal ultrasound data, improving predictive accuracy by **10%**
- Curated and preprocessed **multi-domain clinical imaging datasets**, applying rigorous quantitative evaluation to validate model performance prior to deployment

HealthCare Data Scientist

Jan 2018–Jan 2022

G42 HealthCare

- Improved staffing and capacity decisions by developing **ARIMA/SARIMA and exponential smoothing forecasts** of patient volume, benchmarked against a seasonal-naïve baseline and validated on held-out periods before operational rollout.
- Deployed **predictive analytics pipelines** for patient monitoring, applying **classification and forecasting techniques** to support operational decisions in a highly regulated environment.
- Ingested and integrated **multi-source patient data streams** from across hospital systems into **centralized dashboards**, improving reliability and accessibility for cross-functional care and operations teams.
- Performed **exploratory data analysis (EDA), feature engineering, and statistical validation** on large healthcare datasets to improve model performance.
- Maintained compliance across 750K+ patient records spanning, measured by zero audit findings against health data standards, by conducting recurring data validation and quality audits with engineering and governance partners.

- Developed a predictive maintenance analytics toolbox for Air Booster Compressor (ABC) motors at PETRONAS Gas Berhad Kerteh Utility.
- Built a low-cost, open-source “[IoT-based predictive maintenance system](#)” integrating real-time motor performance data.
- Applied Python-based machine learning models and quantitative analytics to identify anomalies and optimize reliability.
- Research published: “[Predictive maintenance of oil and gas equipment using recurrent neural network](#)”, IEEE ICIAS

PROFESSIONAL EXPERIENCE

- Built **machine-learning-ready data pipelines** supporting **credit risk forecasting models** within a regulated financial services environment, directly informing **risk-based decisioning**.
- Automated **Python reporting workflows**, reducing manual effort by **40%** while strengthening data governance and reproducibility.

SELECTED PUBLICATIONS

- Dounia Marbough, **Tayaba Abbasi**, Fatema Maasmi, Ilhaam Omar, Mazin Debe, Khaled Salah, Raja Jayaraman, Samer Ellahham, [Blockchain for COVID-19: Review, Opportunities and a Trusted Tracking System](#), *Arabian Journal for Science and Engineering*, 2021
- **Tayaba Abbasi**, Lim King Hann and I. Ismail, [Predictive maintenance of oil and gas equipment using recurrent neural network](#), *11th Curtin University Technology, Science and Engineering (CUTSE) International Conference*, 26–28 November 2018, Miri, Malaysia
- **Tayaba Abbasi**, Lim King Hann, N.S. Rosli, I. Ismail and R. Ibrahim, [Development of predictive maintenance interface using multiple linear regression](#), *International Conference on Intelligent and Advanced Systems (ICIAS)*, 13–15 August 2018, Kuala Lumpur, Malaysia

SELECTED PROJECTS

Transformer from Scratch: Encoder–Decoder Architecture in PyTorch

Tools: Python (PyTorch, NumPy, Matplotlib), Jupyter

- Implemented the full encoder–decoder Transformer (*Attention Is All You Need*) from first principles, multi-head attention, sinusoidal positional encoding, causal and padding masks using only `nn.Linear` and tensor ops, verified to `0.00e+00` max absolute deviation against `torch.nn.MultiheadAttention` under copied weights.
- Trained a 932K-parameter model to 99.8% exact-match accuracy on a sequence-reversal task under free-running greedy decoding (3000 steps, ~4.5 min on CPU), by adding pre-norm residuals, Noam LR warmup, label smoothing and weight tying; cross-attention recovered the correct anti-diagonal alignment with no attention supervision.

Modular RAG Pipeline for Graduate NLP Course Preparation

Tools: Python, LangChain, OpenAI SDK, Streamlit, MLflow, YAML

- Architected a fully modular LLM orchestration pipeline integrating OpenAI SDK and LangChain, enabling plug-and-play swapping of retrieval methods (dense, BM25, agentic) and LLM backends via YAML configuration
- Integrated MLflow for experiment tracking across LLM configurations and prompt variants, ensuring reproducibility and systematic evaluation of generation quality.
- Addressed production-level challenges including context window management, token-aware history truncation, and modular component interfaces for fault-tolerant multi-turn conversations

TEACHING EXPERIENCE

- Instructor for the Course - **Generative AI and LLMs: Architecture and Data Preparation** with over **1K+** enrollments and **4.9/5 rating**.
- Creator of **15 Guided Projects and 5 Courses** during my time at Correlation One with over 1K+ enrollments in the field of **Gen AI - Chatbots, Deep Learning and Computer Vision**

CONFERENCE PRESENTATIONS & ACADEMIC ENGAGEMENT (CANADA)

- [Provincial Health Services Authority](#) Data Analytics Summit, Canada (2024), [Predictive Analytics in Public Health: Forecasting Influenza Trends in Canada](#)
- [CAN-CWIC West 2025](#), Simon Fraser University, Vancouver, Canada (2025). [Poster Presentation: Analyzing Housing Affordability in Metro Vancouver](#).